

Mean State of the Tropical Atmosphere

Class Observation and Notes Activity

Name: _____

Date: _____

Purpose. As we examine January and July climatological fields, record what you actually see before we discuss the interpretation. Focus on spatial patterns, seasonal shifts, hemispheric differences, and relationships among variables. As we discuss each panel in class, we will attempt to build a coherent picture of the mean tropical atmosphere.

Note Taking Tips:

1. First pass: write down 2–3 observations without trying to explain them.
2. Compare January and July: what moves, strengthens, weakens, appears, or disappears?
3. After discussion: add the physical interpretation or terminology introduced in class.

1. Surface Winds

Lecture slides 3–4

A. January: What patterns do you notice?

B. July: What patterns do you notice?

C. Seasonal comparison

Over the tropical oceans, the dominant low-level wind direction is:

What changes most dramatically over the North Indian Ocean?

What does the wind field suggest about cross-equatorial flow?

Class notes / interpretation

2. Surface Temperature

Lecture slides 5–6

A. Compare tropical and midlatitude temperature gradients. What do you see?

B. Identify prominent continental warm regions.**January:**

July:

C. Questions to consider

1. Are the strongest seasonal temperature changes over land or ocean? Why might this be?
2. Where would you expect thermally driven surface pressure "heat lows" to occur?

Notes

3. Sea-Level Pressure

Lecture slides 7–8

A. First observations**January****July**

B. Locate the subtropical anticyclones.Where are they found relative to the equator?

How do they change from winter to summer?

Are they zonally symmetric?

C. Connection to surface temperature

How do summer continental heating maxima relate spatially to nearby oceanic pressure systems?

4. Sea-Surface Temperature (SST)**Lecture slides 9–10****A. Mark or describe the following features from the maps.**

Indo-Pacific warm pool: _____

Eastern equatorial Pacific: _____

Tropical vs. midlatitude SST gradients: _____

B. Seasonal comparison

Does the warmest tropical water simply follow the latitude of maximum solar heating? Record what you see.

C. ENSO connection

During a strong El Niño, how would you expect the spatial extent of the warmest equatorial Pacific water to change?

5. Precipitation**Lecture slides 11–14****A. Where is tropical rainfall concentrated? List the main geographical features.**

1. _____
2. _____
3. _____
4. _____

B. Seasonal movement

How does the main tropical rain belt shift between DJF/January and JJA/July?

C. Compare precipitation with SST.

1. Where do high precipitation and high SST generally coincide?
2. Can you find regions with warm SST but relatively little precipitation?
3. What does this tell you about the idea that “warm SST alone causes rainfall”?

D. Identify these features

ITCZ: _____

SPCZ: _____

South Asian monsoon rain: _____

Dry subtropics: _____

6. Zonal-Mean Zonal Wind

Lecture slides 15–16

Sign convention: $u > 0$ is westerly flow; $u < 0$ is easterly flow.**A. Locate the major features in the latitude–height sections.**

Upper-level subtropical westerly jets: _____

Near-surface tropical easterlies: _____

Boreal-summer upper-level tropical easterlies: _____

B. Seasonal comparison of the subtropical jets

Which hemisphere has the stronger and more equatorward jet in January? In July?

C. Dynamical connection

What previously learned relationship helps explain why the upper-level westerly jets are strongest where horizontal temperature gradients are strong?

D. Monsoon connection

What regional circulation contributes to the upper-level easterly maximum just north of the equator during boreal summer?

7. Zonal-Mean Vertical Motion (ω)

Lecture slides 17–18

Recall that pressure vertical velocity is

$$\omega = \frac{Dp}{Dt},$$

so

$$\omega < 0 \Rightarrow \text{ascent}, \quad \omega > 0 \Rightarrow \text{descent}.$$

A. Describe the mean meridional pattern of vertical motion.

B. Approximate locations

Main tropical ascent: _____

Subtropical descent: _____

C. Seasonal asymmetry

How does the latitude and strength of the ascending branch change between January and July?

D. Name the large-scale overturning circulation evident in this plot.

8. Synthesis: Build the Mean Tropical State

Use the observations above to connect the fields rather than treating them independently.

A. Complete the chains below using words or short phrases.

Strong solar heating → → deep convection / rainfall

Tropical ascent → → subtropical descent

Subtropical descent → → dry subtropics

Seasonal continental heating → → monsoon circulation

B. Relationships among observed fields

For each pair, write one sentence describing an important relationship.

1. SST and precipitation:

2. Surface winds and precipitation:

3. Sea-level pressure and surface winds:

4. Vertical motion and precipitation:

5. Surface temperature gradients and upper-level zonal wind:

C. Three defining characteristics of the mean tropical atmosphere

Based on all of the maps, list three features you think are most fundamental.

1.

2.

3.

9. Questions to Carry Forward

These questions will connect today's observations to later course modules.

1. Why are horizontal temperature and pressure gradients weak through much of the tropics?

2. Why is tropical rainfall concentrated into particular convergence zones rather than distributed uniformly over the warm tropical oceans?

3. Why are the subtropics dominated by descent and relatively little precipitation?

4. Why are the January and July tropical circulations not simply mirror images of one another?

10. Free Class Notes
